



**NYU**

Center for Urban  
Science + Progress

DEC 12 JUNYI LI

# NYC Breathe: Revolutionizing Allergy Management by NYC DOHMH

# A NEWYORKER'S STORY



- **Severe symptoms of pollen allergy significantly impact daily life.**
- **Patients are unwilling to invest time and effort in seeking medical attention.**
- **The lack of street-level geographical information in alert limits patient assistance.**

# ALLERGY FACTS

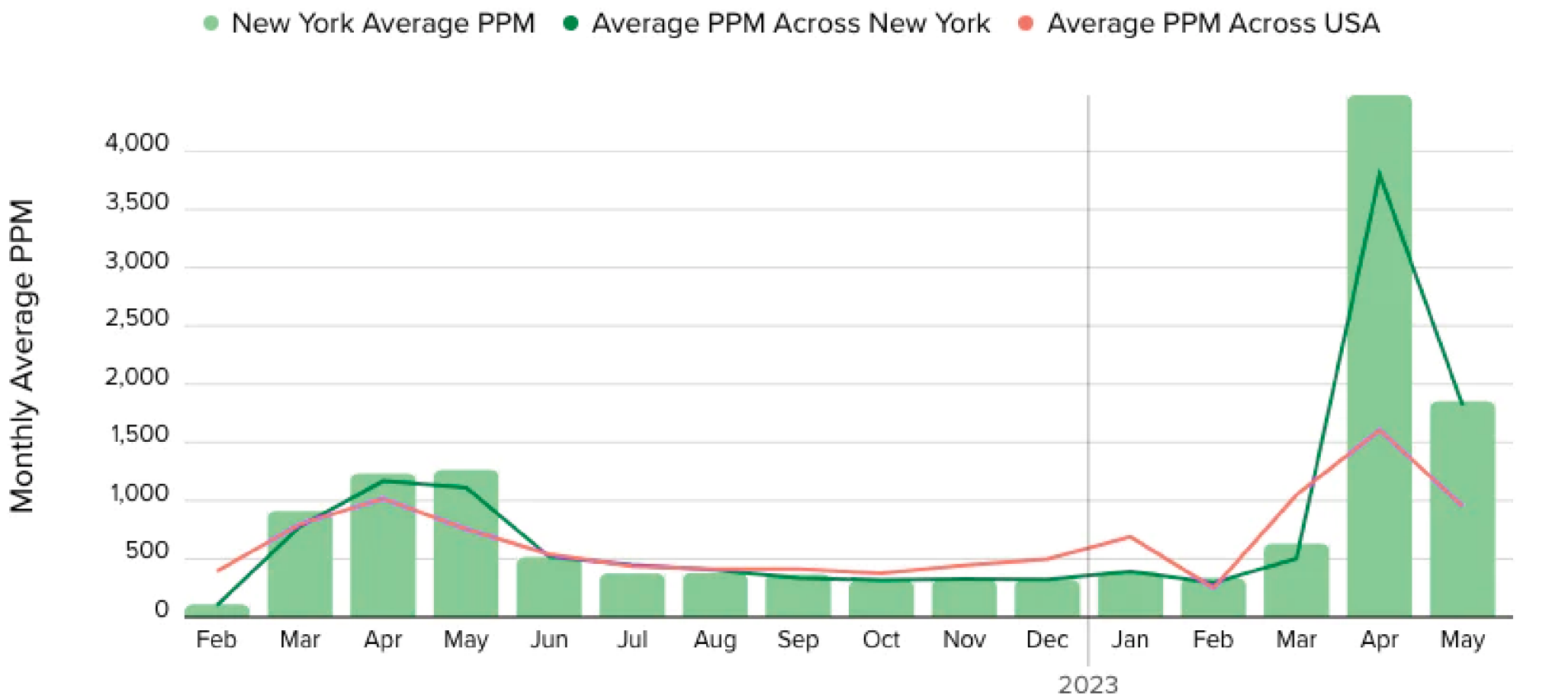
**81 MILLION**  
PEOPLE IN THE U.S.  
DIAGNOSED WITH HAY FEVER

**26%**  
OF ADULTS

**19%**  
OF CHILDREN

## New York, NY Pollen PPM

February 2022 – May 2023

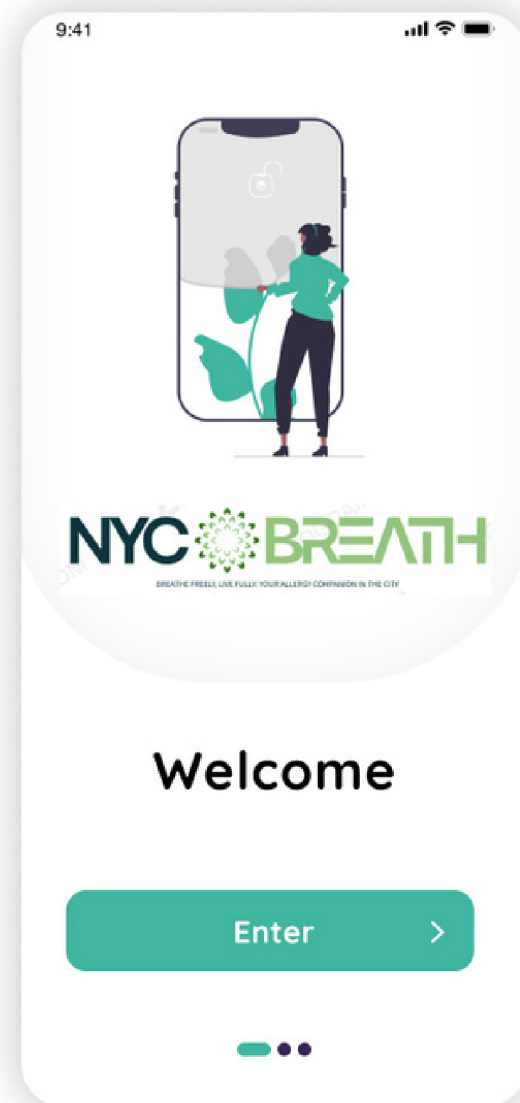


wyndly.com | **w** WYNDLY

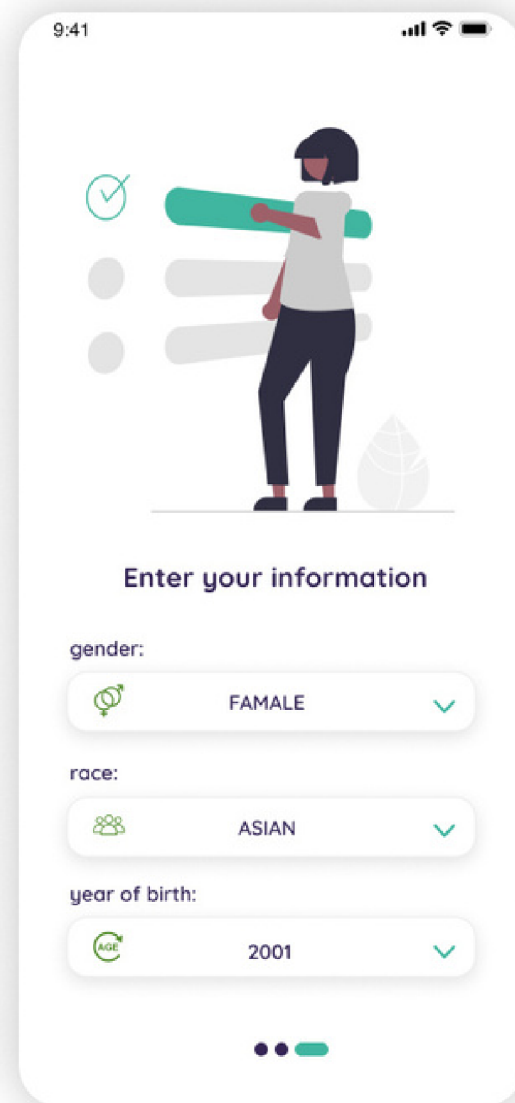
**0 DATASET IN NYC**

# NYC BREATHE

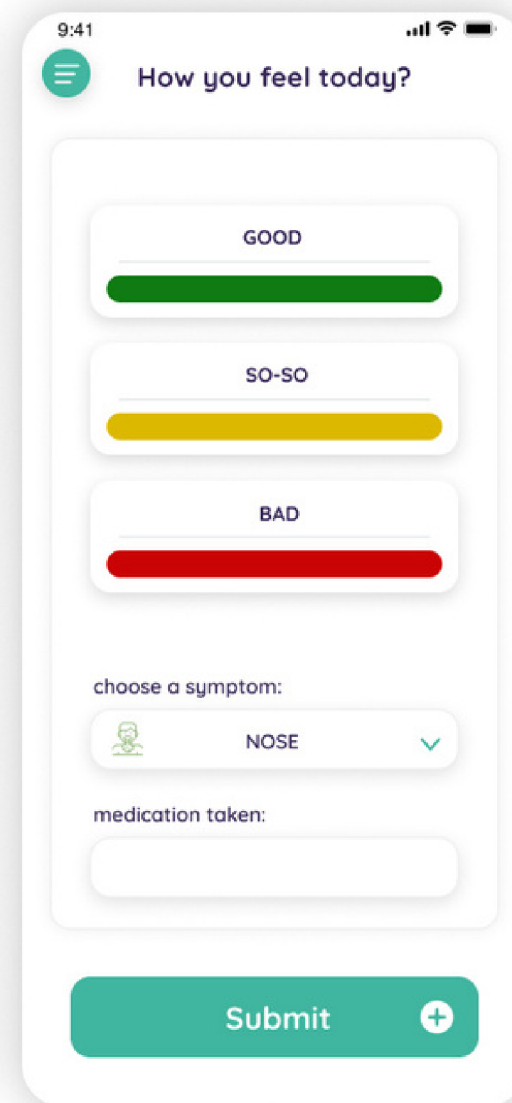
Breathe Freely, Live Fully: Your Allergy Companion in the City



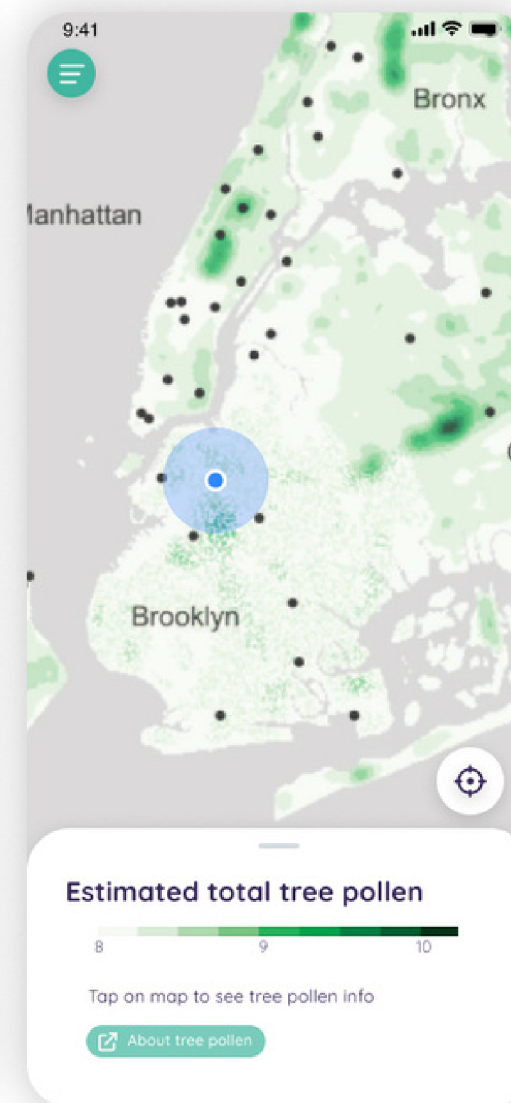
WELCOME



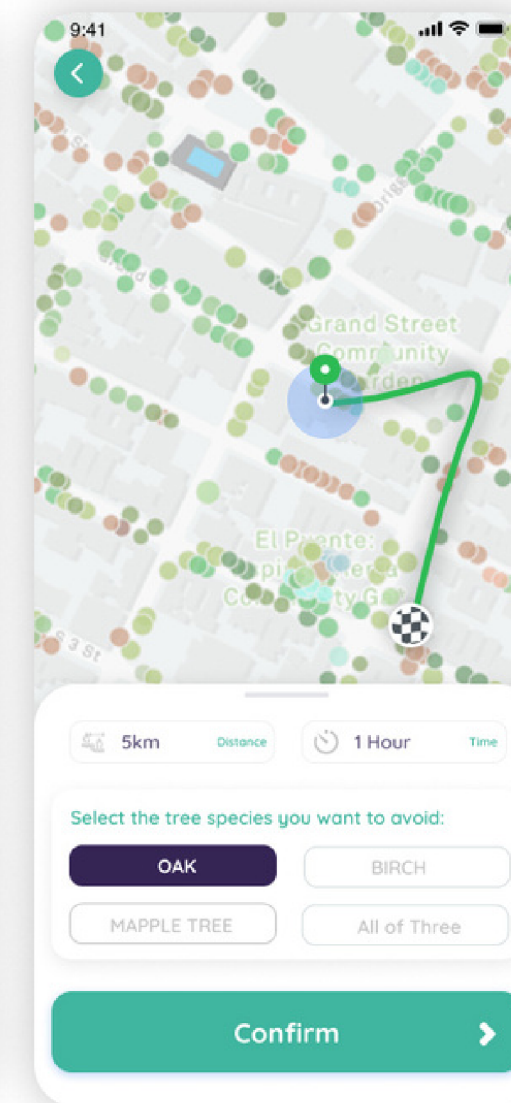
INFO COLLECT



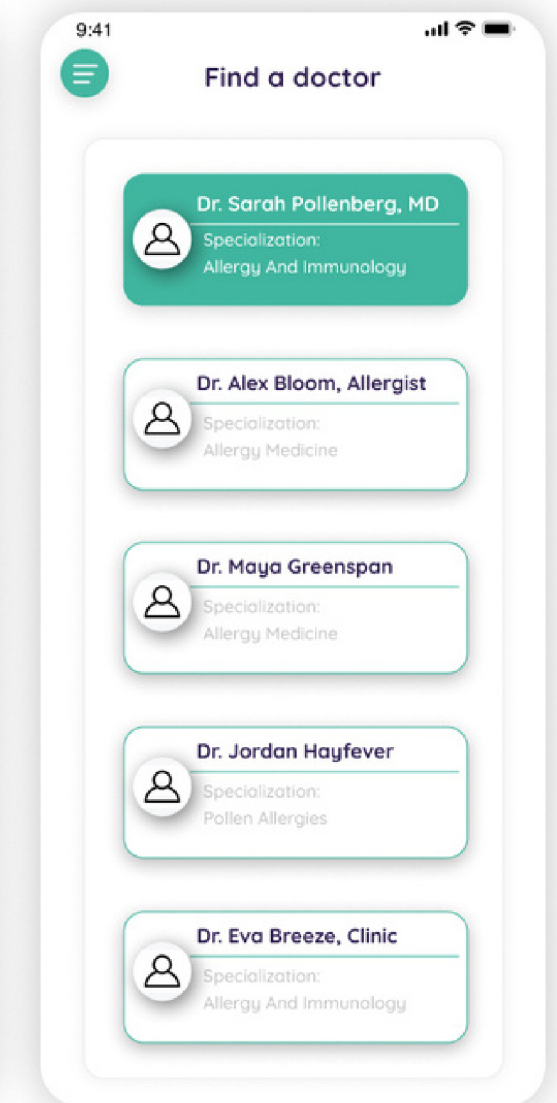
SELF REPORT



POLLEN MAP



NAVIGATION



CONTACT DOCTOR



# THREE PHASE

1

## ORGANIZATIONAL REFORM

- Reforms in organizational structure, culture, budget, technology, collaboration, and equity

2

## DATA COLLECTION BY SELF-REPORTING

- Design and publish a test phase application
- Ask participants to submit wellness and symptoms

3

## LONG TERM STRATEGY

- Add alert and medical advice function to the app
- Policies Development and data management



# PARTNERS

1

## NYC PARK DEPARTMENT

- Provide NYC tree map and add tree blooming data
- Push hypoallergenic green space planning

2

## NYC HEALTH + HOSPITAL

- Provide medical resources

3

## POLLEN SENSE LLC.

- Provide Automated Airborne-Particle Sensors



# CHALLENGE

1

## **COSTS AND LIMITATION OF SENSOR**

- Counting is labor-intensive
- Automated counting is not accurate
- Efficiently offer pollen details by integrating treemap and flowering data

2

## **DATA PRIVACY**

- Applying careful de-identified data methods
- prevent the misuse of health data by pharmaceutical companies

3

## **LIMITATIONS OF SELF-REPORT**

- Self-selection biases, and attrition
- Build trust and enhanced advocacy



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# NYC Breathe

a community-driven solution to tackle pollen allergies head-on

**Thank you!**